

Intro to AI and Data Science



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● Agenda

- Why to learn AI or ML ?
- What is actually AI?
- Narrow vs. General AI.
- Why this Booming? Just a trend?
- Why is AI Now ?
- Data Never Sleeps
- What is ML?
- ML vs. Traditional Programming.
- AI vs. ML vs. DL vs. DS.
- Data Science Life Cycle.
- Types of ML.
- Applications of ML.
- Job Roles.



Why to learn AI or ML ?



“

“AI is the new electricity”

*“Artificial intelligence is everywhere!
Just as electricity revolutionized lives
100 years ago, AI is changing our lives
completely today”*

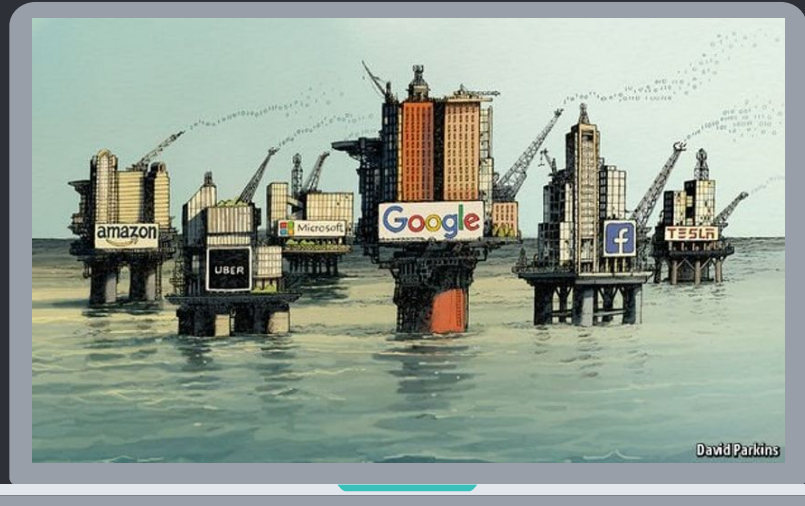


Andrew Ng

The world's most valuable resource is no longer oil, but data.

The Economist

(May 2017)



Article link :

<https://www.economist.com/leaders/2017/05/06/the-worlds-most-valuable-resource-is-no-longer-oil-but-data>

10 Signs You're a Technology Dinosaur in a High-Tech World



(July 2020)



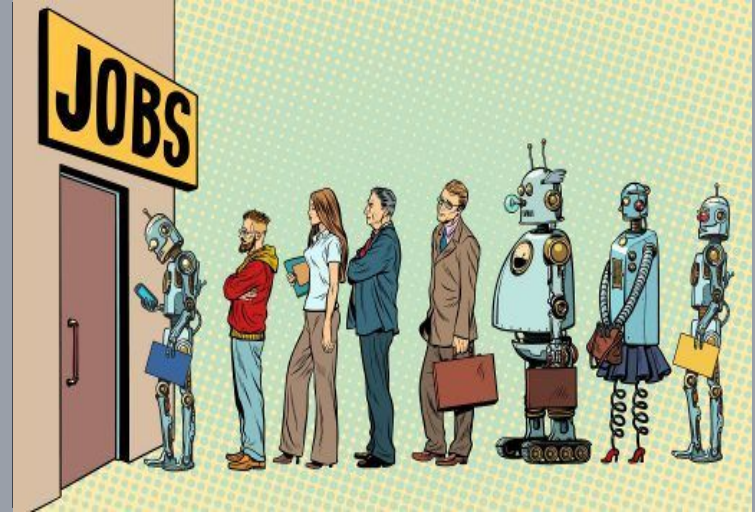
Article link :

<https://www.suddenlysenior.com/10-signs-youre-a-dinosaur/>

The future of employment: can we be optimistic about the impact of new technologies?



(Mar 2018)



Article link :
<https://epthinktank.eu/2018/03/13/the-future-of-employment-can-we-be-optimistic-about-the-impact-of-new-technologies/>



What is actually AI?



“

Can machines think?



Alan Turing

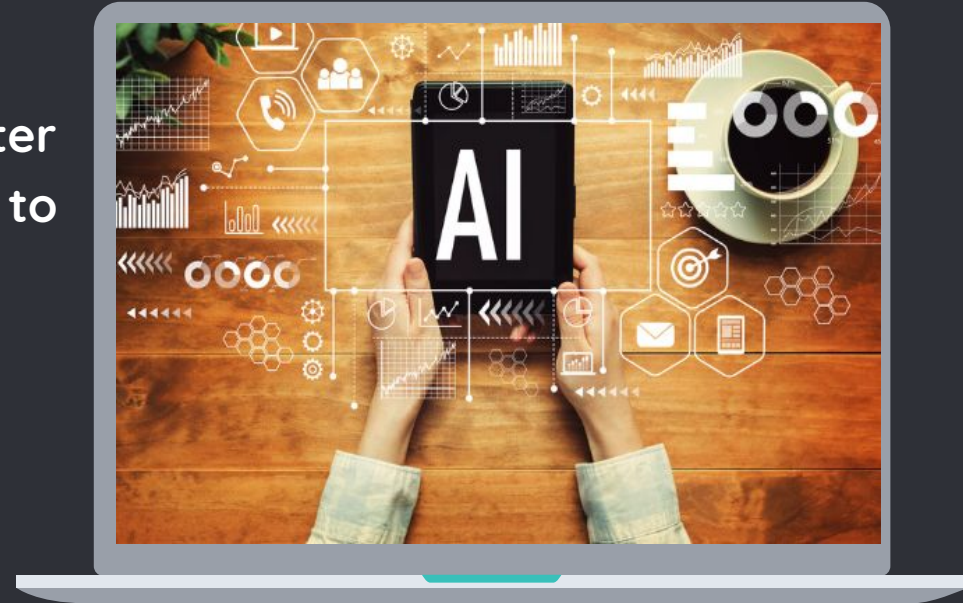
1912-1954

Simulation of Natural Intelligence to mimic human actions *

** Informal Definition*



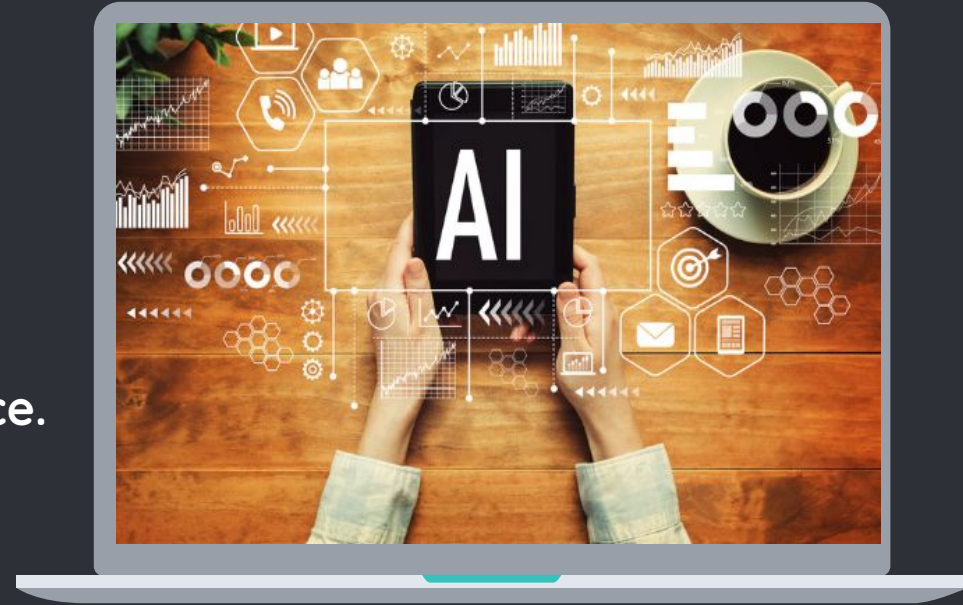
The ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings.



Article link :
<https://www.britannica.com/technology/artificial-intelligence>

The ability to:

- Reason,
- discover meaning,
- generalize, or
- learn from past experience.



Article link :
<https://www.britannica.com/technology/artificial-intelligence>



Narrow vs. General AI

- Narrow vs. General AI

- **Narrow AI (Weak)**

Machines perform Specific tasks with human intelligence or may be better.



- **General AI (Strong)**

Machines perform General tasks with human intelligence.



- Narrow vs. General AI

Super AI

*Machines perform General tasks
better than human intelligence
(outperform human Intelligence)*

Do you think it is possible ? 🤔

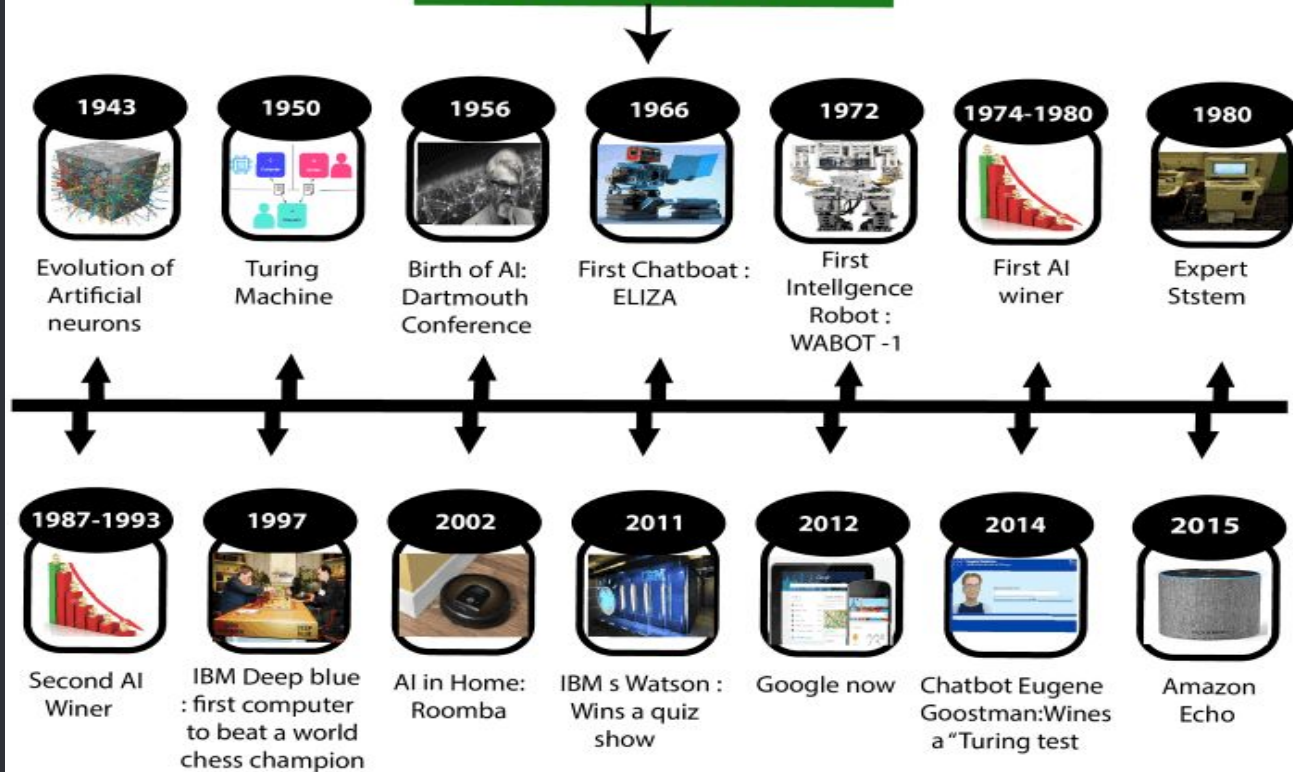




Why this Booming? Just a trend?



History of AI



Source: <https://www.javatpoint.com/history-of-artificial-intelligence>

- Why this Booming? Just a trend?

○ The first AI winter (1974-1980)

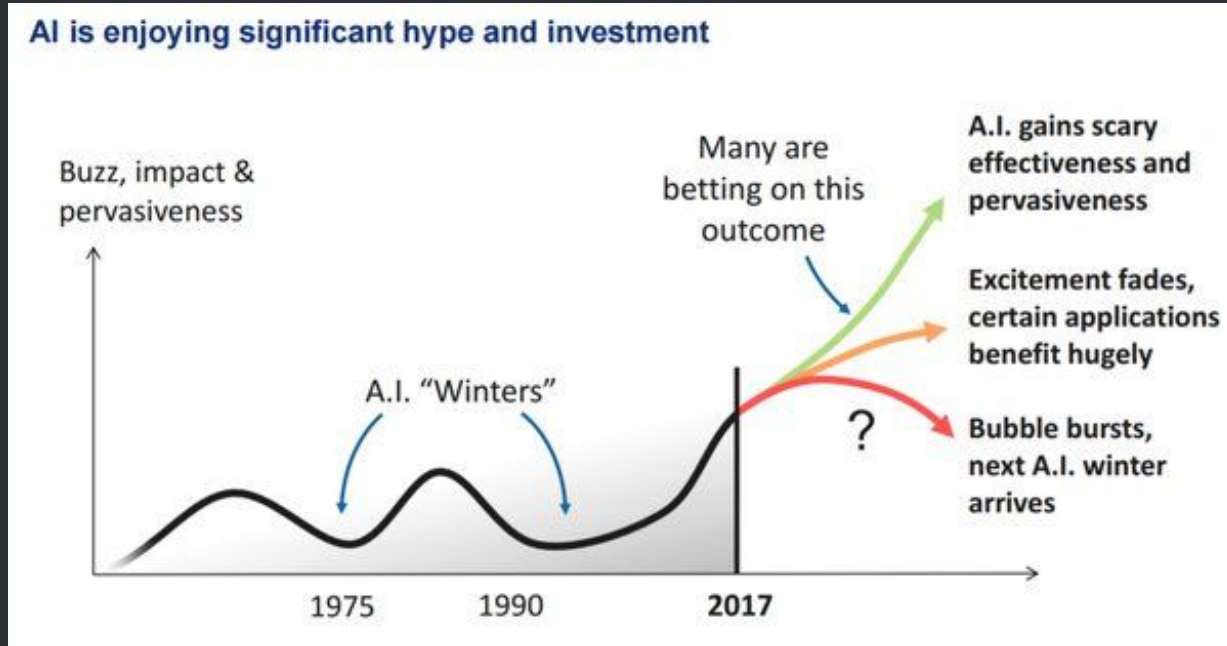
- *AI winter refers to the time period where computer scientists dealt with a severe shortage of funding from government for AI researches.*
- *During AI winters, an interest of publicity on artificial intelligence was decreased.*

- Why this Booming? Just a trend?

○ The second AI winter (1987-1993)

- *AI Again Investors and government stopped in funding for AI research as due to high cost but not efficient result.*
- *Business drive Research*
- Business Wise : What is in it for me ? 🤔

- Why this Booming? Just a trend?



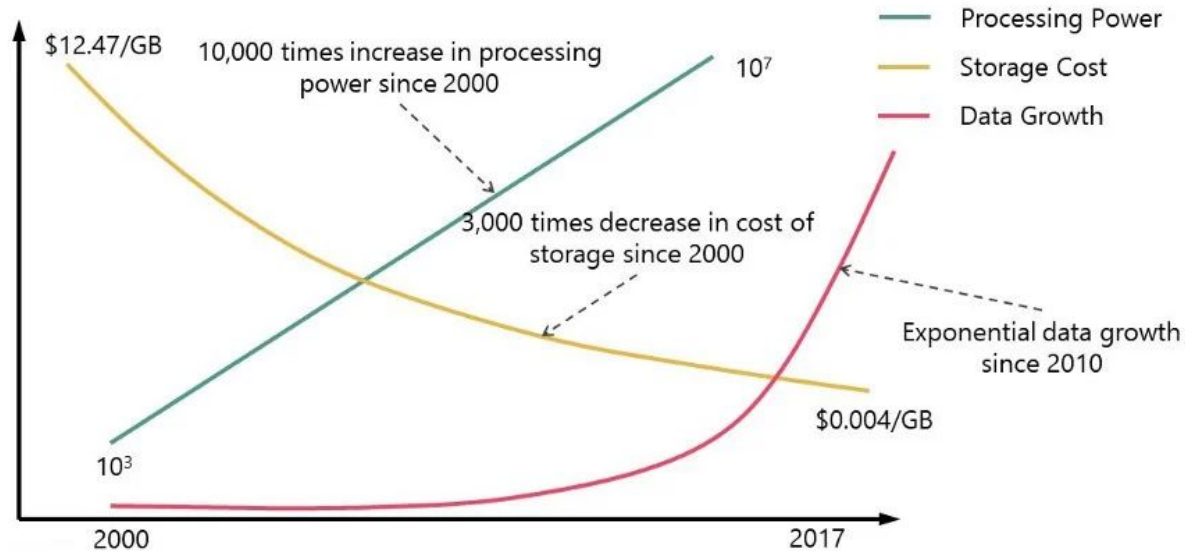
Source: [Link from Quora](#)



Why is AI Now ?



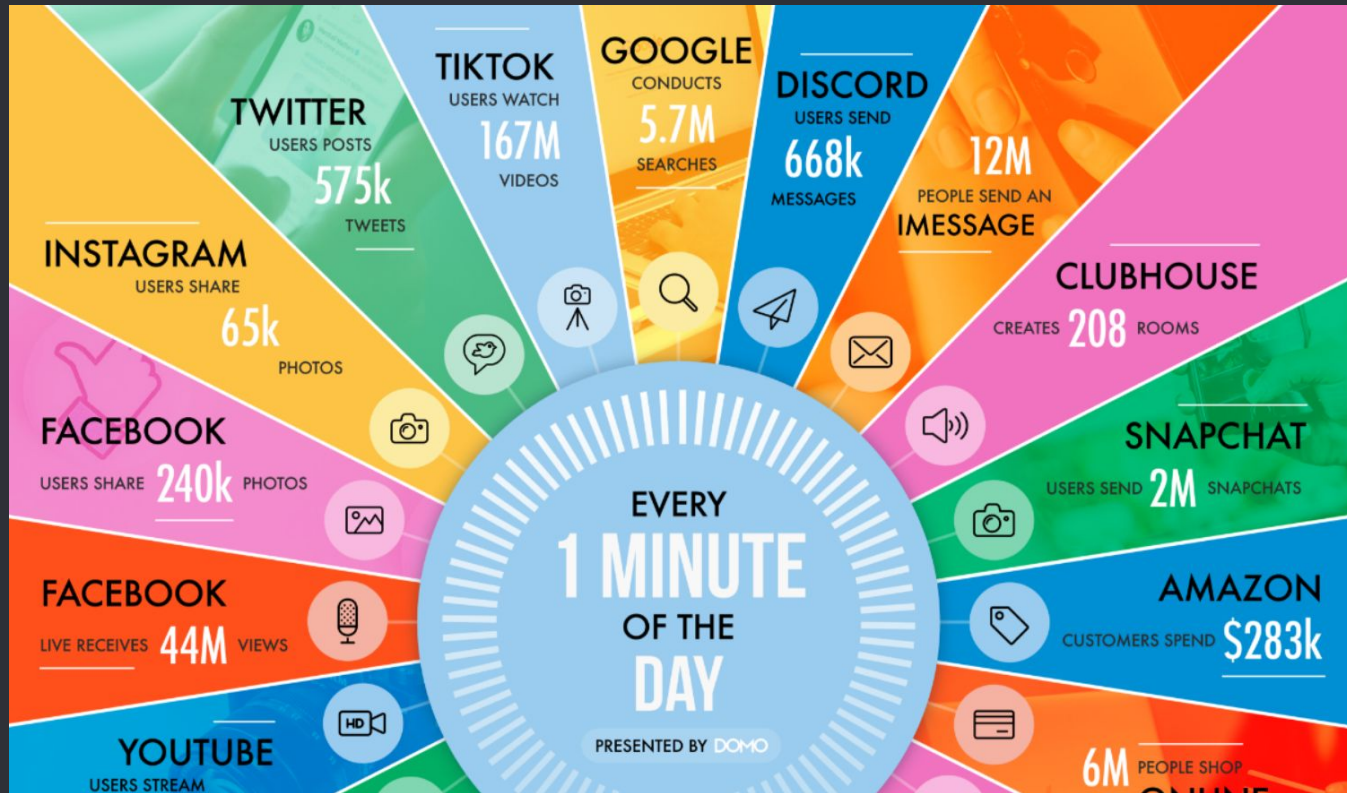
Why AI is on the Rise



Source: <https://www.eteam.io/blog/computer-engineering-automated>



Data Never Sleeps



Data Never Sleeps

Source: <https://www.domo.com/learn/infographic/data-never-sleeps-9>



What is ML?

What is Machine Learning?

“

Field of study that gives computers the ability to learn without being explicitly programmed”



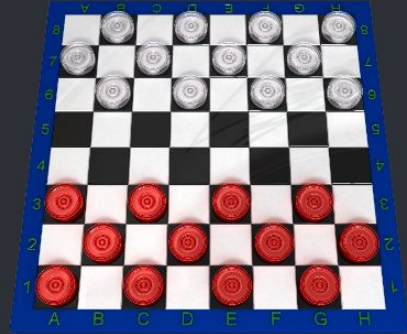
Arthur Samuel

1959

What is Machine Learning?

“

A checkers program played maybe tens of thousands of games against itself, learned over time what are good or bad positions by trying to get a good and avoid bad positions.



Learn by Experience

What is Machine Learning?

“

“A computer program is said to learn from experience (E) with respect to some task (T) and some performance measure (P), if its performance improves with experience”



Tom Mitchell

1997

What is Machine Learning?

“

- *Learning by Experience*
- *Learning from Data*
- *Pattern Recognition*



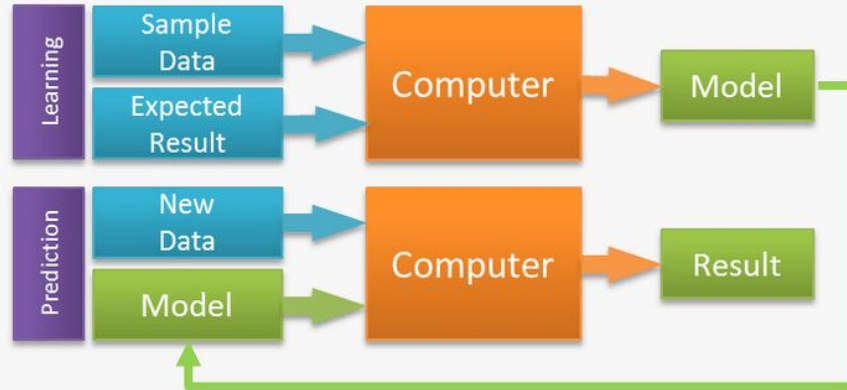


ML vs. Traditional Programming

Traditional modeling:



Machine Learning:

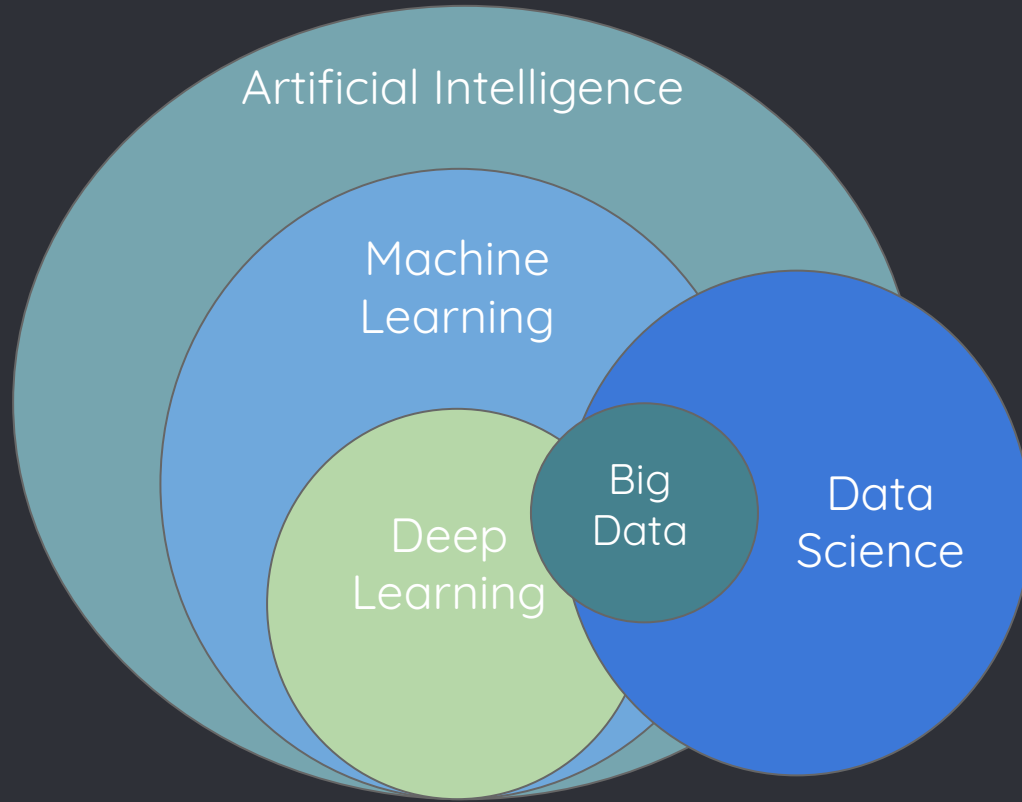


ML vs. Traditional Programming

Source: https://www.researchgate.net/publication/343079524_Detection_of_Offensive_Language_in_Social_Media_Posts



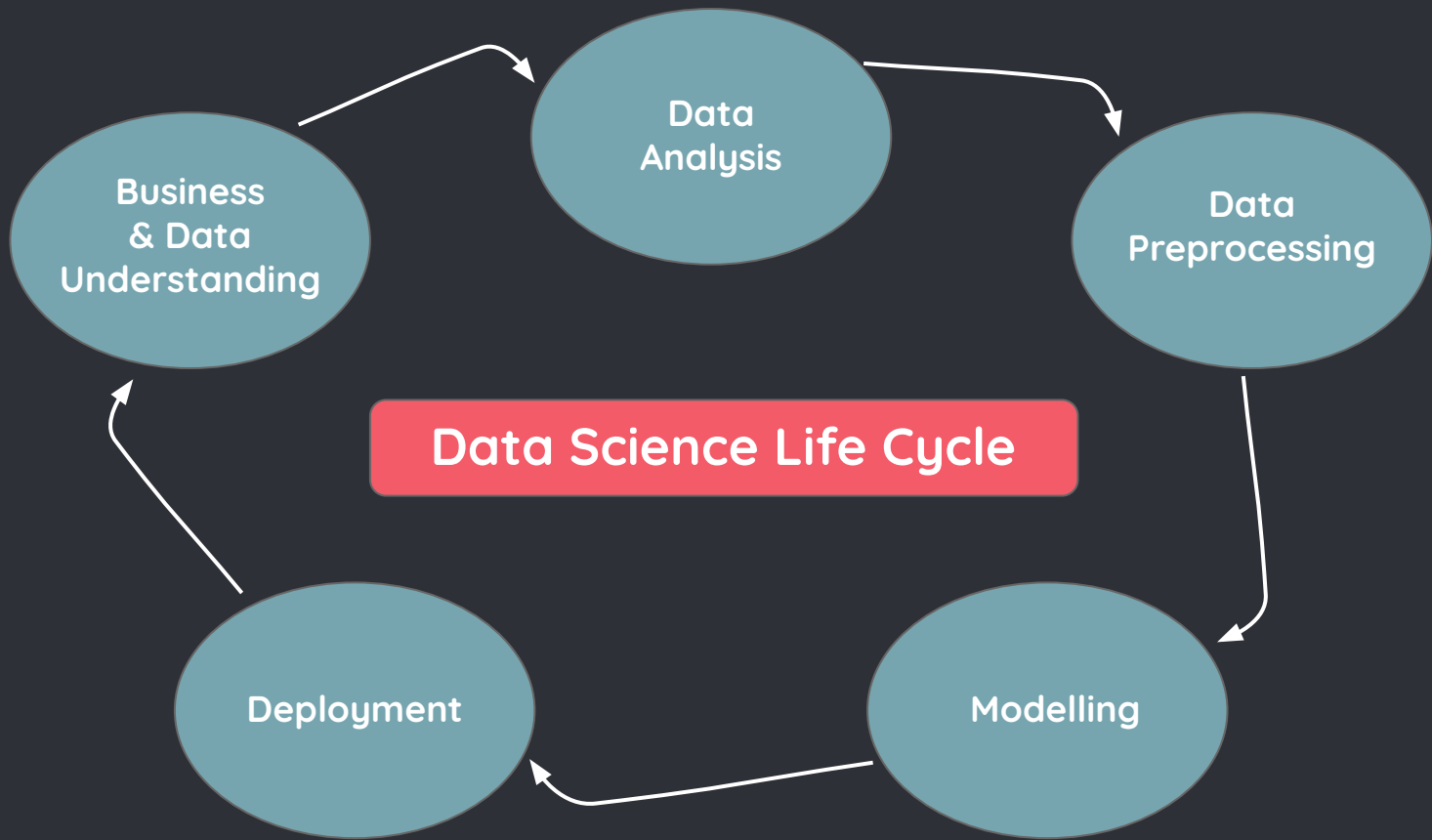
AI vs. ML vs. DL vs. DS



AI vs. ML vs. DL vs. DS



Data Science Life Cycle



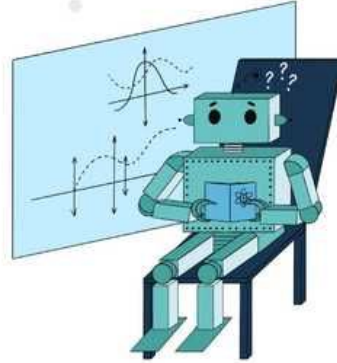


Types of ML

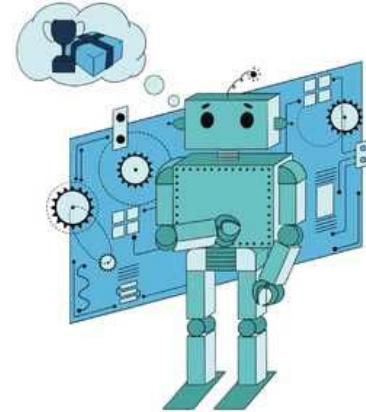
Types of Machine Learning



Supervised Learning



Unsupervised Learning

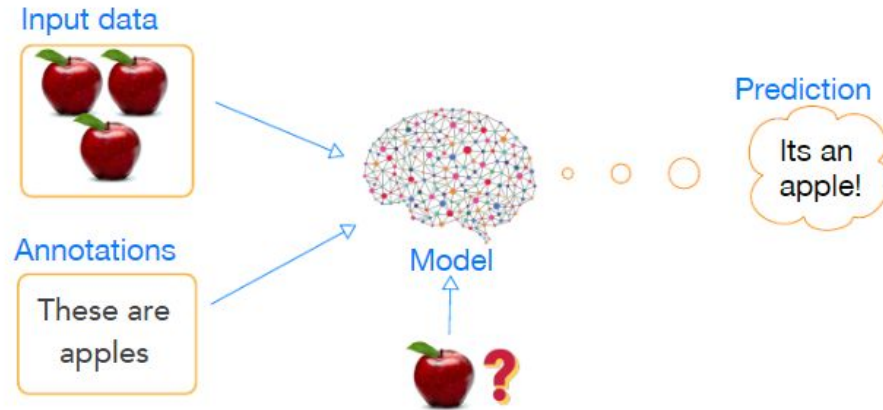


Reinforcement Learning

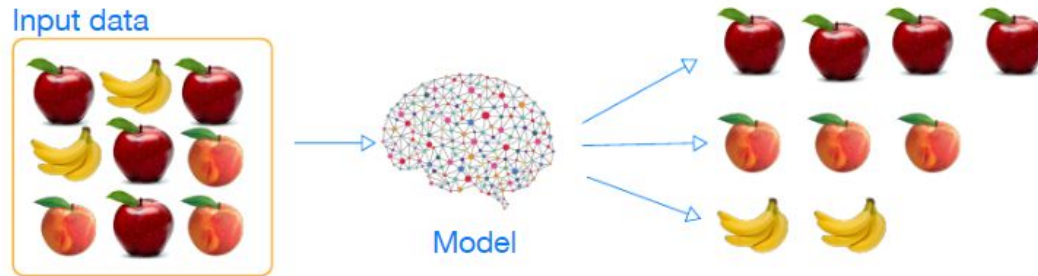
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Source: <https://blog.autify.com/machine-learning-in-software-testing>

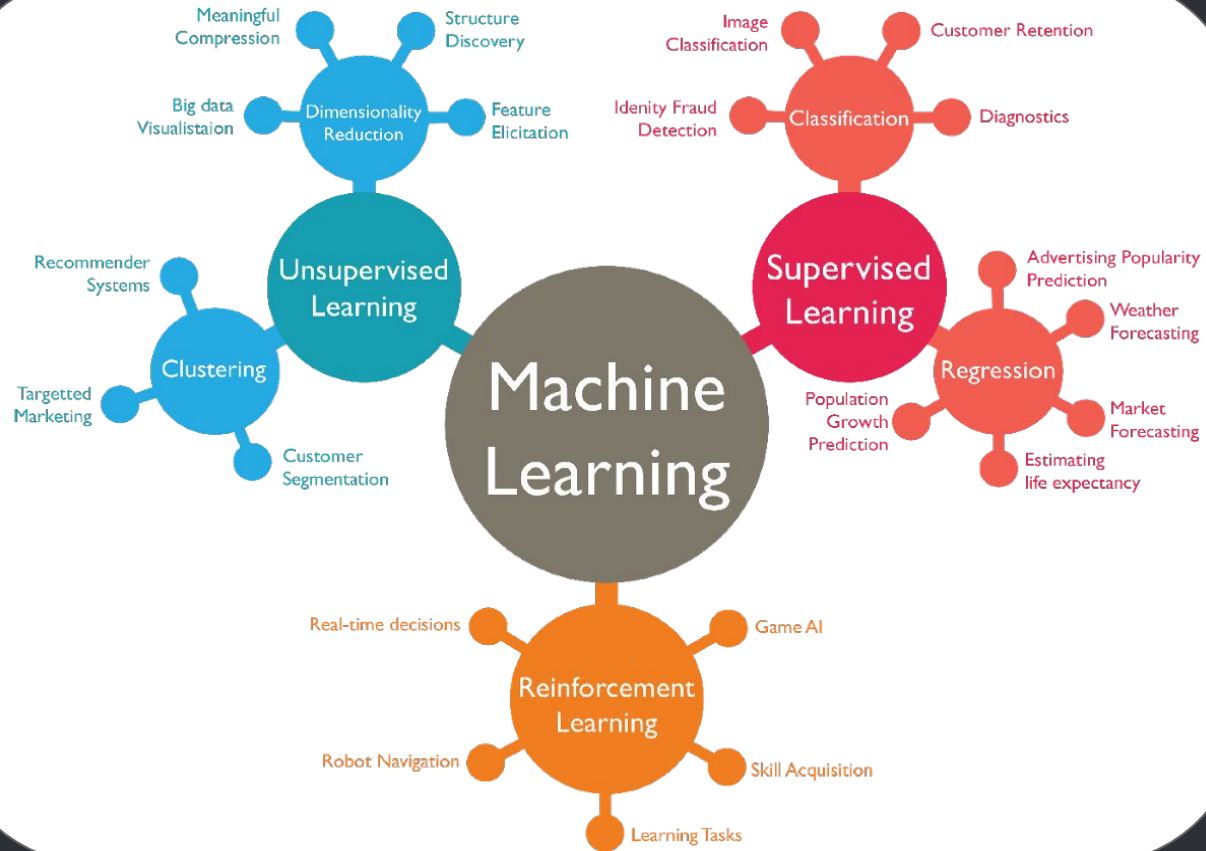
supervised learning



unsupervised learning



Source: [researchgate](https://www.researchgate.net)



Source: <https://towardsdatascience.com/machine-learning-types-2-c1291d4f04b1>



Applications of ML

Some Applications of Machine Learning

- *Car Price Prediction*
- *Bike Sharing Demand*



Regression

- *Ad Click Prediction*
- *Dogs vs. Cats Image Classification*
- *Mall Customers Segmentation*



Classification

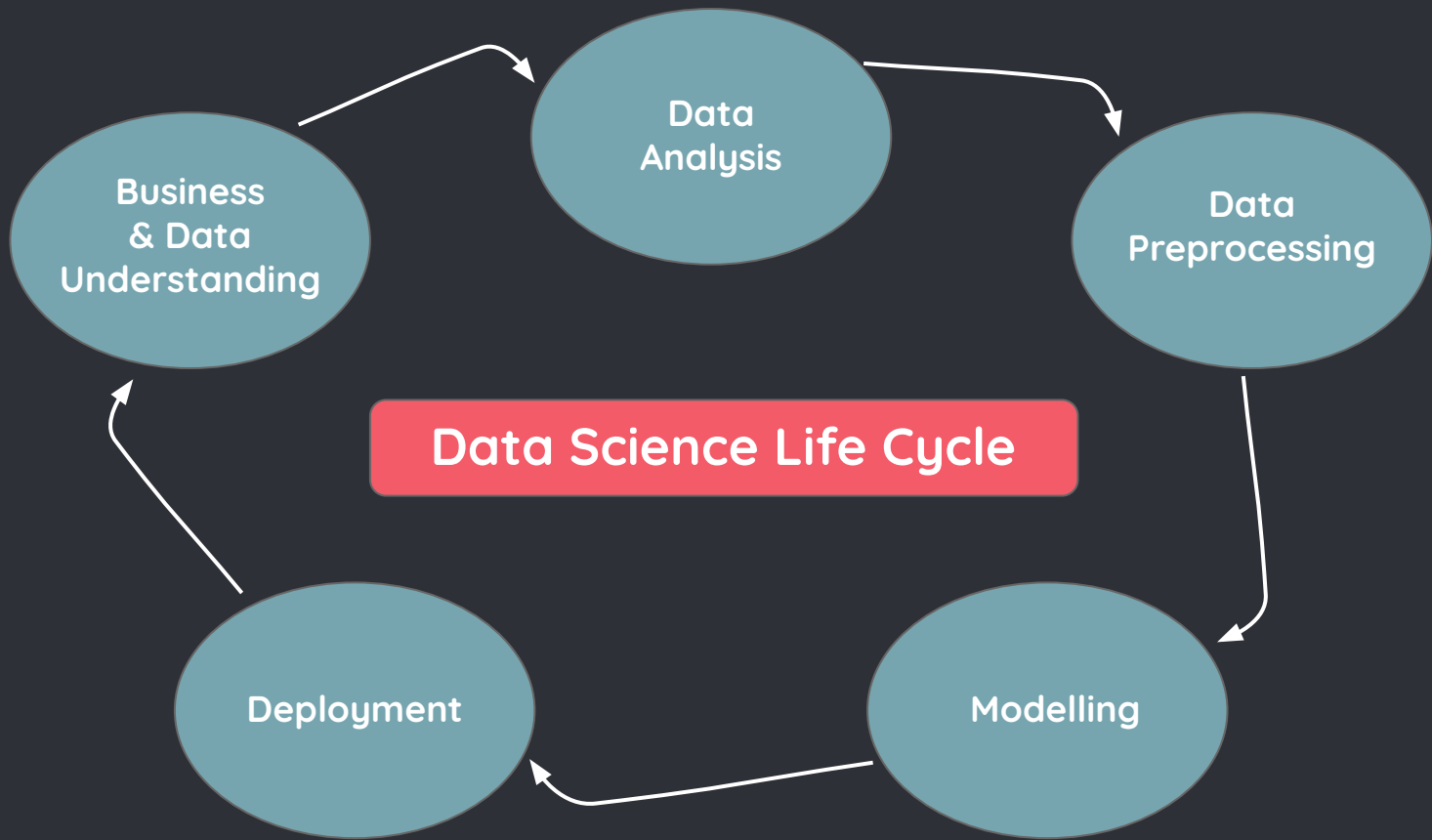


Clustering





Job Roles



Data Engineer



Main Task

Build and maintain data pipeline

Main Skills

- Coding
- Database
- Big Data
- Cloud

Data Analyst



Main Task

Explore, Visualize
and Communicate

Main Skills

- Coding
- Statistics
- Dashboards
- Business

ML Engineer



Main Task

Build, tune and deploy ML model

Main Skills

- Coding
- Statistics
- ML Algorithms
- Cloud

Data Scientist



Main Task

Analyze, tune and drive whole cycle

Main Skills

- Coding
- Statistics
- ML Algorithms
- Domain Knowledge



Done

KEEP
GOING